WORKED SOLUTION

# STEP II 1997, Question 12

Probability and Statistics • Binomial Parity Probability

**STEP II 1997, Question 12**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q12

## The key idea

Track only whether the message is currently *right* or *wrong*. Each mishearing flips it, so the message that comes back to Albert is right exactly when an even number of the whispers went wrong. Picking out the even terms of a binomial expansion is what $(a+b)^n + (a-b)^n$ does, since the odd terms cancel. For the modified game, notice that after one failure Albert passes on a *wrong* message, so from then on each round needs an *odd* number of mishearings.

## Solution

### Three players

With $n$ players there are $n$ whispers: Albert to Bertha, Bertha to the next player, and so on round to Zebedee, then Zebedee back to Albert. Each whisper is heard correctly with probability $p$, independently. Whether Albert hears the truth depends only on how many whispers were misheard, and not on what the coin showed.

With three players there are three whispers, each correct with probability $\frac{2}{3}$. The message that reaches Albert is right if no whisper goes wrong, or if exactly two do (the second error undoes the first). So

$$\Pr(\text{won}) = \left(\tfrac{2}{3}\right)^3 + 3\left(\tfrac{1}{3}\right)^2\left(\tfrac{2}{3}\right) = \frac{8}{27} + \frac{6}{27} = \frac{14}{27}.$$

### $n$ players

Let $q = 1 - p$. The number of misheard whispers has the $\mathrm{B}(n, q)$ distribution, and the game is won when it is even:

$$P = \sum_{k \text{ even}} \binom{n}{k} q^k p^{n-k}.$$

Now

$$(p+q)^n + (p-q)^n = \sum_{k=0}^{n} \binom{n}{k} p^{n-k} q^k (1 + (-1)^k) = 2 \sum_{k \text{ even}} \binom{n}{k} p^{n-k} q^k,$$

because the odd-$k$ terms cancel and the even-$k$ terms appear twice. Since $p + q = 1$ and $p - q = 2p - 1$,

$$\boxed{P = \tfrac{1}{2}(1 + (2p-1)^n).}$$

For $n = 3$, $p = \frac{2}{3}$ this gives $\frac{1}{2}(1 + \frac{1}{27}) = \frac{14}{27}$, agreeing with the first part.

#### The modified game

Let $N$ be the number of times Albert whispers to Bertha. Call each pass of the message all the way round the circle a *round*, so every round begins with Albert whispering.

**Ask yourself** After a failed round, is the next round just a fresh copy of the first?

No. In the first round Albert sends the correct result. If he hears it wrongly from Zebedee, he passes on what he *heard*, which is wrong. Starting from a wrong message, the round ends with Albert hearing the truth exactly when an *odd* number of whispers go wrong. That has probability $1 - P$. If the round fails again, Albert has once more heard (and passes on) a wrong message, so every later round is like this one. The rounds are independent, because the hearings are.

So the first round succeeds with probability $P$, and each later round succeeds with probability $1 - P$ and fails with probability $P$. Hence $\Pr(N = 1) = P$, and for $k \geq 2$

$$\Pr(N = k) = \underbrace{(1-P)}_{\text{round 1 fails}} \underbrace{P^{k-2}}_{\text{rounds 2,...,}k-1\text{ fail}} \underbrace{(1-P)}_{\text{round } k \text{ wins}} = (1-P)^2 P^{k-2}.$$

Assume $0 < p < 1$, so that $0 < P < 1$. Then, using the given sum with $x = P$,

$$\sum_{k=2}^{\infty} k P^{k-2} = \frac{1}{P}\left(\sum_{k=1}^{\infty} k P^{k-1} - 1\right) = \frac{1}{P}\left(\frac{1}{(1-P)^2} - 1\right),$$

and so

$$\begin{aligned} \mathrm{E}(N) &= P + (1-P)^2 \cdot \frac{1}{P}\left(\frac{1}{(1-P)^2} - 1\right) \\ &= P + \frac{1 - (1-P)^2}{P} \\ &= P + (2 - P) = 2. \end{aligned}$$

The expected number of times Albert whispers to Bertha is 2, whatever the number of players and whatever $p$ is.

There is a quicker way to see this. Given that the first round fails, the number of further rounds is geometric with success probability $1 - P$, so its mean is $1/(1-P)$. Hence $\mathrm{E}(N) = 1 + (1-P) \cdot \frac{1}{1-P} = 2$.

(At the edges: if $p = 1$, or if $p = 0$ with $n$ even, then $P = 1$ and Albert whispers just once. If $p = 0$ with $n$ odd, then $P = 0$ and the second round is certain to succeed, so $N = 2$.)

## Checks

- Listing all $2^n$ patterns of right and wrong hearings with exact fractions reproduces $\frac{1}{2}(1 + (2p-1)^n)$ for $n \leq 8$ and several values of $p$, including $\Pr(\text{won}) = \frac{14}{27}$ for three players.
- $p = \frac{1}{2}$ gives $P = \frac{1}{2}$: the message is pure noise, as it should be.
- Simulating $2 \times 10^5$ modified games with $(n, p) = (3, \frac{2}{3})$, $(5, 0.9)$ and $(4, 0.3)$ gave average whisper counts 2.001, 1.996 and 1.994.

## Common slips

- Counting $n - 1$ whispers instead of $n$: the last whisper, from Zebedee back to Albert, counts too.
- Treating every round of the modified game as winning with probability $P$. That gives $1/P$, which is wrong, because after a failure Albert passes on a wrong message.

## Beyond STEP

The right/wrong state of the message is a two-state Markov chain, and $2p - 1$ is the second eigenvalue of its transition matrix. That is why $(2p-1)^n$ controls how fast the message forgets the coin. The same model, the binary symmetric channel, is the starting point of information theory and error-correcting codes. Grimmett and Welsh, *Probability: An Introduction*, covers generating functions and Markov chains at this level.

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